

Student: Jonatan Rabjerg**Instructor:** Andreas Basse-O'Connor**Assignment:** MyLab Math - Uge 5**Date:** 1/15/24**Course:** Calculus Beta

Use the Integral Test to determine whether the series shown below converges or diverges. Be sure to check that the conditions of the Integral Test are satisfied.

$$\sum_{k=6}^{\infty} \frac{1}{3e^{10k}}$$

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a

positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)dx$ both converge or both diverge.

The function $f(x) = \frac{1}{3e^{10x}}$ will be used for the Integral Test. Note that f has the following properties.

The function $f(x)$ is continuous for $x \geq 6$.

The function $f(x)$ is positive for $x \geq 6$.

The function $f(x)$ is a decreasing function for $x \geq 6$.

The function $f(x)$ has the property that $a_k = f(k)$ for $k = 6, 7, 8, \dots$

Since all four conditions are satisfied, the Integral Test can be used to determine if the sequence $\sum_{k=6}^{\infty} \frac{1}{3e^{10k}}$ converges or diverges.

To determine if the integral $\int_6^{\infty} \frac{dx}{3e^{10x}}$ converges or diverges, first find the value of the corresponding indefinite integral.

$$\int \frac{dx}{3e^{10x}} = -\frac{e^{-10x}}{30} + C$$

Use the Fundamental Theorem of Calculus and the indefinite integral found above to evaluate the definite integral.

$$\int_6^{\infty} \frac{dx}{3e^{10x}} = \lim_{b \rightarrow \infty} \int_6^b \frac{dx}{3e^{10x}}$$

Use the definition of an improper integral.

$$= \lim_{b \rightarrow \infty} \left[-\frac{e^{-10x}}{30} \right]_6^b$$

Evaluate the integral.

$$= \lim_{b \rightarrow \infty} \left[\left(-\frac{e^{-10b}}{30} \right) - \left(-\frac{e^{-10(6)}}{30} \right) \right]$$

Simplify.

Now find the limit.

$$\lim_{b \rightarrow \infty} \left[\left(-\frac{e^{-10b}}{30} \right) - \left(-\frac{e^{-10(6)}}{30} \right) \right] = \frac{e^{-60}}{30}$$

$$\text{Thus, } \int_6^{\infty} \frac{dx}{3e^{10x}} = \frac{e^{-60}}{30}.$$

Since $\int_6^{\infty} \frac{dx}{3e^{10x}}$ converges, by the Integral Test, the series $\sum_{k=6}^{\infty} \frac{1}{3e^{10k}}$ also converges.